THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE CHAPTER TEST SET B

Chapter 1: **Exploring Substances: Acidic, Basic, and Neutral** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Blue litmus turns ___ in an acidic solution. 1

(a) red **(b)** green **(c)** yellow **(d)** blue

2. Sugar solution is: 1

(a) a strong acid **(b)** basic **(c)** neutral **(d)** acidic

3. The acid present in curd is: 1

(a) oxalic acid **(b)** lactic acid **(c)** citric acid **(d)** acetic acid

4. Turmeric paper turns ___ with a basic substance. 1

(a) colourless **(b)** reddish-brown **(c)** green **(d)** blue

5. The products of neutralisation are: 1

(a) acid and base **(b)** salt and water
(c) oxygen and hydrogen **(d)** only salt

6. To neutralise basic (alkaline) soil, one can add: 1

(a) soap **(b)** organic matter (compost)
(c) baking soda **(d)** lime

7. Vanilla essence smell in a basic solution: 1

(a) disappears **(b)** turns sour **(c)** stays the same **(d)** gets stronger

8. Washing soda is used in soaps because it is: 1

(a) neutral **(b)** sour **(c)** basic **(d)** acidic

9. A solution does not change the colour of either litmus. It is: 1

(a) basic **(b)** a strong acid **(c)** acidic **(d)** neutral

10. Which of the following is a neutral substance? 1

(a) Lemon juice **(b)** Vinegar **(c)** Lime water **(d)** Distilled water

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. Why is baking soda applied on an ant bite? 2

12. Why should we not taste unknown substances in the laboratory? 2

13. Name the acids present in vinegar, curd and lemon. 2

14. What happens to turmeric paper when soap solution is dropped on it? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why is excess use of fertilisers harmful to soil? 3

16. What is the importance of acids and bases in the digestive system? 3

17. Differentiate between acids and bases on the basis of taste, touch and litmus. 3

OR

A student tests four solutions: P turns blue litmus red; Q turns red litmus blue; R no change; S turns blue litmus red. Classify. 3

18. Why is toothpaste basic? How does it protect teeth? 3

OR

Explain why we should not use litmus paper made with turmeric for acids. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Explain the role of indicators in classifying substances. 5
(ii) Complete the table for lemon juice, soap, sugar using litmus.
(iii) Why should the test be repeated with red and blue litmus?

OR

(i) Why is soil acidity a problem? 5
(ii) How can farmers correct acidic and basic soil?
(iii) What is the role of neutralisation?

20. **Case study:** A river near a factory has fish dying. The water tests acidic. 5
(i) Cause?
(ii) Test?
(iii) Treatment?
(iv) Why does neutralisation help?

OR

(i) Name acids found in five foods. 5
(ii) State their taste.
(iii) Why is it not safe to taste them all?

END OF QUESTION PAPER — Total: 40 Marks

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